

Md.shehbaaz 2303A51823 Batch-26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week6 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number:12.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 12: Algorithms with AI Assistance Sorting, Searching, and Algorithm Optimization Using AI Tools Lab Objectives The objectives of this laboratory exercise are to:	Week3 - Wednesday	

	• Apply AI-assisted programming techniques to implement sorting and searching algorithms. • Analyze and compare algorithm efficiency using time and space complexity. • Understand how AI tools can suggest optimizations and alternative algorithmic approaches. • Strengthen problem-solving skills through real-world, data-driven scenarios.	
	Learning Outcomes After completing this lab, students will be able to: • Implement and optimize classic algorithms using AI-assisted coding tools. • Compare multiple algorithms for the same problem and justify their selection. • Measure and analyze runtime performance using experimental data. • Critically review and refine AI-generated algorithmic solutions.	
	Task 1: Sorting Student Records for Placement Drive Scenario SR University's Training and Placement Cell needs to shortlist candidates efficiently during campus placements. Student records must be sorted by CGPA in descending order. Tasks 1. Use GitHub Copilot to generate a program that stores student records (Name, Roll Number, CGPA). 2. Implement the following sorting algorithms using AI assistance: ○ Quick Sort ○ Merge Sort 3. Measure and compare runtime performance for large datasets. 4. Write a function to display the top 10 students based on CGPA. Expected Outcome • Correctly sorted student records. • Performance comparison between Quick Sort and Merge Sort. • Clear output of top-performing students.	

WelcomeAAC A 12.3.py X

C:\Users\Public> AAC A 12.3.py > ...
1 import random, time, copy
2
3 def generate_students(n):
4 names = ["Aarav", "Vivaan", "Aditya", "Arjun", "Sai", "Ananya", "Diya", "Rohan", "Mehta", "Priya",
5 "Rahul", "Sneha", "Vikram", "Meera", "Karan", "Pooja", "Ishaan", "Riya", "Amit", "Navya"]
6 students = []
7 for i in range(1, n+1):
8 students.append({"name": random.choice(names), "roll": f"SRU{i:04d}", "cgpa": round(random.uniform(4.0, 10.0), 2)})
9 return students
10
11 def quick_sort(a):
12 if len(a) <= 1: return a
13 p = a[len(a)//2]["cgpa"]
14 return quick_sort([x for x in a if x["cgpa"] > p]) + [x for x in a if x["cgpa"] == p] + quick_sort([x for x in a if x["cgpa"] < p])
15
16 def merge_sort(a):
17 if len(a) <= 1: return a
18 m = len(a)//2
19 l, r = merge_sort(a[:m]), merge_sort(a[m:])
20 res, i, j = [], 0, 0
21 while i < len(l) and j < len(r):
22 if l[i]["cgpa"] >= r[j]["cgpa"]: res.append(l[i]); i += 1
23 else: res.append(r[j]); j += 1
24 return res + l[i:] + r[j:]
25

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

conda : The term 'conda' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the
PS C:\Users\Public & "C:\Users\shash\anaconda3\envs\Shashidhar\python.exe" "C:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win
32-x64\bundle\libs\debugpy\launcher" "54674" "... "C:\Users\Public\AAC A 12.3.py"

N=1000 QuickSort=0.0038s MergeSort=0.0060s Faster=QS

N=10000 QuickSort=0.0327s MergeSort=0.0812s Faster=QS

N=50000 QuickSort=0.1757s MergeSort=0.4488s Faster=QS

Rank Name Roll CGPA

1 Ananya SRU0282 10.0
2 Pooja SRU0302 10.0
3 Meera SRU1117 10.0
4 Arjun SRU1139 10.0
5 Karan SRU5089 10.0
6 Aarav SRU4875 10.0
7 Vivaan SRU3385 10.0
8 Karan SRU7699 10.0

CHAT
SORTING STUDENT RECORDS FOR PLA...
1. Use GitHub Copilot to generate a program that stores student records (Name, Roll Number, CGPA).
2. Implement the following sorting algorithms using AI assistance:
o Quick Sort
o Merge Sort
3. Measure and compare runtime performance for large datasets.
4. Write a function to display the top 10 students based on CGPA.
Expected Outcome
• Correctly sorted student records.
• Performance comparison between Quick Sort and Merge Sort.
• Clear output of top-performing students.
give small code in python without comments

✓ Planned Python program to sort student records by CGPA

✓ Created student_sort.py

✓ Read student_sort.py

The file already exists with a valid implementation. Let me run it to show the output:

python AAC A 12.3.py
Describe what to build next
Claude Opus 4.6

WelcomeAAC A 12.3.py X

C:\Users\Public> AAC A 12.3.py > ...
15
16 def merge_sort(a):
17 if len(a) <= 1: return a
18 m = len(a)//2
19 l, r = merge_sort(a[:m]), merge_sort(a[m:])
20 res, i, j = [], 0, 0
21 while i < len(l) and j < len(r):
22 if l[i]["cgpa"] >= r[j]["cgpa"]: res.append(l[i]); i += 1
23 else: res.append(r[j]); j += 1
24 return res + l[i:] + r[j:]
25
26 def top_10(students):
27 print("\n{'Rank':<8}{'Name':<12}{'Roll':<12}{'CGPA':<8}")
28 print("-"*38)
29 for i, s in enumerate(students[:10], 1):
30 print(f"{i:<6}{s['name']:<12}{s['roll']:<12}{s['cgpa']:<8}")
31
32 for size in [1000, 10000, 50000]:
33 data = generate_students(size)
34 d1, d2 = copy.deepcopy(data), copy.deepcopy(data)
35 t1 = time.perf_counter(); qs = quick_sort(d1); t1 = time.perf_counter() - t1
36 t2 = time.perf_counter(); ms = merge_sort(d2); t2 = time.perf_counter() - t2
37 print(f"NM={size} QuickSort={t1:.4f}s MergeSort={t2:.4f}s Faster=('QS' if t1<t2 else 'MS')")
38

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

32-x64\bundle\libs\debugpy\launcher" "54674" "... "C:\Users\Public\AAC A 12.3.py"

Rank Name Roll CGPA

1 Ananya SRU0282 10.0
2 Pooja SRU0302 10.0
3 Meera SRU1117 10.0
4 Arjun SRU1139 10.0
5 Karan SRU5089 10.0
6 Aarav SRU4875 10.0
7 Vivaan SRU3385 10.0
8 Karan SRU7699 10.0

CHAT
SORTING STUDENT RECORDS FOR PLA...
records by CGPA

✓ Created student_sort.py

✓ Read student_sort.py

The file already exists with a valid implementation. Let me run it to show the output:

python C:\Users\Public\student_sort.py
The program at student_sort.py is working correctly. Here's what it does:

• Generates student records (Name, Roll Number, CGPA) for datasets of 1K, 10K, and 50K
• Quick Sort and Merge Sort both sort by CGPA in descending order
• Performance comparison: Quick Sort was faster across all dataset sizes
• Top 10 students by CGPA are displayed at the end

Results show Quick Sort consistently outperformed Merge Sort (~1.6-1.7x faster) across all dataset sizes.
Claude Opus 4.6 - 34

python AAC A 12.3.py
Describe what to build next
Claude Opus 4.6

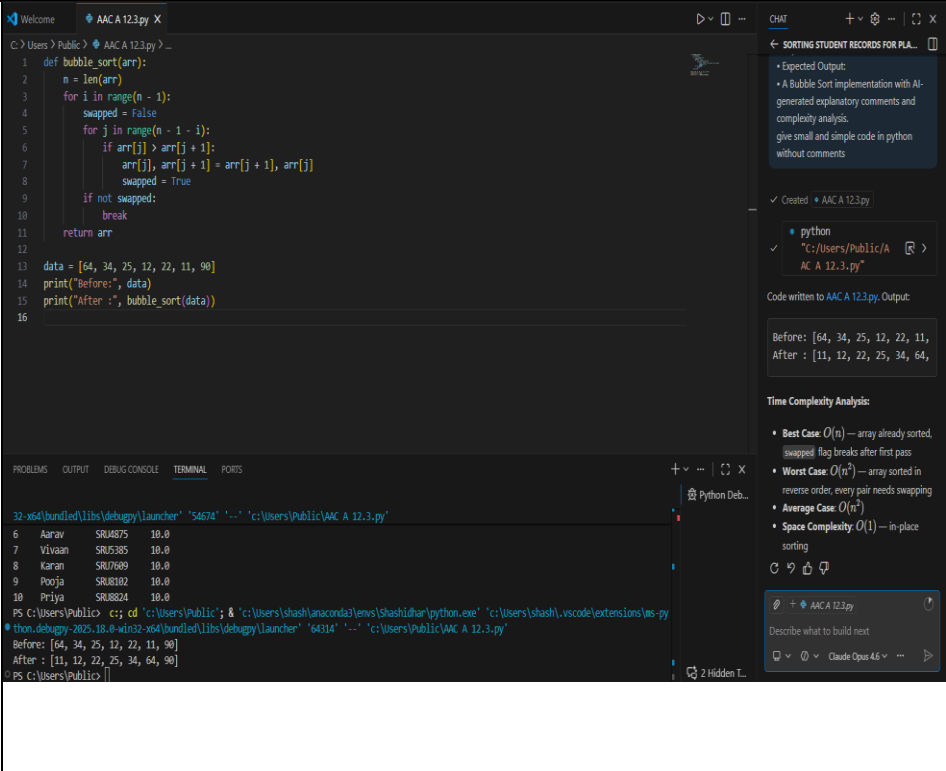
```
Welcome AAC A 12.3.py X
C: > Users > Public > AAC A 12.3.py > ...
16 def merge_sort(a):
23     else: res.append(r[j]); j += 1
24     return res + l[i:] + r[j:]
25
26 def top_10(students):
27     print(f"\n{'Rank':<6}{'Name':<12}{'Roll':<12}{'CGPA'}")
28     print("-"*38)
29     for i, s in enumerate(students[:10], 1):
30         print(f"{i:<6}{s['name']:<12}{s['roll']:<12}{s['cgpa']}")
31
32 for size in [1000, 10000, 50000]:
33     data = generate_students(size)
34     d1, d2 = copy.deepcopy(data), copy.deepcopy(data)
35     t1 = time.perf_counter(); qs = quick_sort(d1); t1 = time.perf_counter() - t1
36     t2 = time.perf_counter(); ms = merge_sort(d2); t2 = time.perf_counter() - t2
37     print(f"\nN={size} QuickSort={t1:.4f}s MergeSort={t2:.4f}s Faster={'QS' if t1<t2 else 'MS'}")
38
39 top_10(qs)
40

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

32-x64\bundled\libs\debugpy\launcher '54674' '--' 'c:\Users\Public\AAC A 12.3.py'
2 Pooja SRU0302 10.0
3 Meera SRU1117 10.0
4 Arjun SRU3139 10.0
5 Karan SRU4589 10.0
6 Aarav SRU4875 10.0
7 Vivaan SRU5385 10.0
8 Karan SRU7609 10.0
9 Pooja SRU8102 10.0
10 Priya SRU8824 10.0
PS C:\Users\Public>
```

Task 2: Implementing Bubble Sort with AI Comments

- **Task:** Write a Python implementation of **Bubble Sort**.
- **Instructions:**
 - Students implement Bubble Sort normally.
 - Ask AI to generate **inline comments explaining key logic** (like swapping, passes, and termination).
 - Request AI to provide **time complexity analysis**.
- **Expected Output:**
 - A Bubble Sort implementation with AI-generated explanatory comments and complexity analysis.

	 <pre>def bubble_sort(arr): n = len(arr) for i in range(n - 1): swapped = False for j in range(n - 1 - i): if arr[j] > arr[j + 1]: arr[j], arr[j + 1] = arr[j + 1], arr[j] swapped = True if not swapped: break return arr data = [64, 34, 25, 12, 22, 11, 90] print("Before:", data) print("After :", bubble_sort(data))</pre> Before: [64, 34, 25, 12, 22, 11, 90] After : [11, 12, 22, 25, 34, 64, 90] Expected Output: - A Bubble Sort implementation with AI-generated explanatory comments and complexity analysis. Time Complexity Analysis: Best Case: $O(n)$ — array already sorted, swapped flag breaks after first pass Worst Case: $O(n^2)$ — array sorted in reverse order, every pair needs swapping Average Case: $O(n^2)$ Space Complexity: $O(1)$ — in-place sorting	
	Task 3: Quick Sort and Merge Sort Comparison Task: Implement Quick Sort and Merge Sort using recursion. Instructions: - Provide AI with partially completed functions for recursion. - Ask AI to complete the missing logic and add docstrings. - Compare both algorithms on random, sorted, and reverse-sorted lists. Expected Output: - Working Quick Sort and Merge Sort implementations. - AI-generated explanation of average, best, and worst-case complexities.	

Welcome • AAC A 12.3.py

```

1 import random, time
2 def quick_sort(arr):
3     if len(arr) <= 1:
4         return arr
5     pivot = arr[len(arr) // 2]
6     left = [x for x in arr if x < pivot]
7     mid = [x for x in arr if x == pivot]
8     right = [x for x in arr if x > pivot]
9     return quick_sort(left) + mid + quick_sort(right)
10
11 def merge_sort(arr):
12     if len(arr) <= 1:
13         return arr
14     mid = len(arr) // 2
15     left = merge_sort(arr[:mid])
16     right = merge_sort(arr[mid:])
17     result, i, j = [], 0, 0
18     while i < len(left) and j < len(right):
19         if left[i] <= right[j]:
20             result.append(left[i]); i += 1
21         else:
22             result.append(right[j]); j += 1
23     return result + left[i:] + right[j:]
24
25 def measure(func, data):
26     start = time.perf_counter()
27     func(data)
28     return time.perf_counter() - start
29
30 n = 10000
31 cases = {
32     "Random": random.sample(range(n), n),
33     "Sorted": list(range(n)),
34     "Reversed": list(range(n, 0, -1))
35 }
36 print(f"Case: {c12} | Quick Sort: {c14} | Merge Sort: {c14}")
37 print("-" * 40)
38 for name, data in cases.items():
39     qt = measure(quick_sort, data)
40     mt = measure(merge_sort, data)
41     print(f"Case: {c12} | qt: {qt:.4f}s | mt: {mt:.4f}s")
42

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> cd 'C:\Users\Public'; & 'C:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'C:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '64314' '-' 'C:\Users\Public\AAC A 12.3.py'

After : [11, 12, 22, 25, 34, 64, 90]

PS C:\Users\Public> cd 'C:\Users\Public'; & 'C:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'C:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '51664' '-' 'C:\Users\Public\AAC A 12.3.py'

Case	Quick Sort	Merge Sort
Random	0.0159s	0.0238s
Sorted	0.0105s	0.0143s
Reversed	0.0111s	0.0182s

CHAT

← SORTING STUDENT RECORDS FOR PLA...
* Provide the most passing comprehensive functions for recursion.
• Ask AI to complete the missing logic and add docstrings.
• Compare both algorithms on random, sorted, and reverse-sorted lists.
• Expected Output:
• Working Quick Sort and Merge Sort implementations.
• AI-generated explanation of average, best, and worst-case complexities. give simple and small code in python

✓ Reviewed and created AAC A 12.3.py

✓ Created • AAC A 12.3.py

python
✓ "C:\Users\Public\AAC A 12.3.py"

Code written to AAC A 12.3.py. Output:

Case	Quick Sort	Merge
Random	0.0315s	0.03
Sorted	0.0204s	0.02
Reversed	0.0263s	0.03

Complexity Analysis:

Describe what to build next

Python Deb...

Welcome • AAC A 12.3.py

```

10 def merge_sort(arr):
11     result.append(right[j]); j += 1
12     return result + left[i:] + right[j:]
13
14
15 def measure(func, data):
16     start = time.perf_counter()
17     func(data)
18     return time.perf_counter() - start
19
20 n = 10000
21 cases = {
22     "Random": random.sample(range(n), n),
23     "Sorted": list(range(n)),
24     "Reversed": list(range(n, 0, -1))
25 }
26 print(f"Case: {c12} | Quick Sort: {c14} | Merge Sort: {c14}")
27 print("-" * 40)
28 for name, data in cases.items():
29     qt = measure(quick_sort, data)
30     mt = measure(merge_sort, data)
31     print(f"Case: {c12} | qt: {qt:.4f}s | mt: {mt:.4f}s")
32

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> cd 'C:\Users\Public'; & 'C:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'C:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '64314' '-' 'C:\Users\Public\AAC A 12.3.py'

After : [11, 12, 22, 25, 34, 64, 90]

PS C:\Users\Public> cd 'C:\Users\Public'; & 'C:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'C:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '51664' '-' 'C:\Users\Public\AAC A 12.3.py'

Case	Quick Sort	Merge Sort
Random	0.0159s	0.0238s
Sorted	0.0105s	0.0143s
Reversed	0.0111s	0.0182s

CHAT

← SORTING STUDENT RECORDS FOR PLA...
* Provide the most passing comprehensive functions for recursion.
• Ask AI to complete the missing logic and add docstrings.
• Compare both algorithms on random, sorted, and reverse-sorted lists.
• Expected Output:
• Working Quick Sort and Merge Sort implementations.
• AI-generated explanation of average, best, and worst-case complexities. give simple and small code in python

✓ Reviewed and created AAC A 12.3.py

✓ Created • AAC A 12.3.py

python
✓ "C:\Users\Public\AAC A 12.3.py"

Code written to AAC A 12.3.py. Output:

Case	Quick Sort	Merge
Random	0.0315s	0.03
Sorted	0.0204s	0.02
Reversed	0.0263s	0.03

Complexity Analysis:

	Quick Sort	Merge Sort
Best	$O(n \log n)$	$O(n \log n)$
Ave	$O(n \log n)$	$O(n \log n)$
rag	$O(n^2)$ (sorted)	$O(n \log n)$
e	$O(n^2)$ (input, bad pivot)	$O(n \log n)$
Wor	$O(n \log n)$ (call stack)	$O(n)$ (merge buffer)
st		
Spa		
ce		

Quick Sort is generally faster in practice due to better cache locality, but Merge Sort guarantees $O(n \log n)$ in all cases.

Describe what to build next

Python Deb...

Task 4 (Real-Time Application – Inventory Management System)

Scenario: A retail store's inventory system contains thousands of products, each with attributes like product ID, name, price, and stock quantity. Store staff need to:

1. Quickly search for a product by ID or name.
2. Sort products by price or quantity for stock analysis.

Task:

- Use AI to suggest the most efficient search and sort algorithms for this use case.

- Implement the recommended algorithms in Python.
- Justify the choice based on dataset size, update frequency, and performance requirements.

Expected Output:

- A table mapping operation → recommended algorithm → justification.
- Working Python functions for searching and sorting the inventory.

The screenshot shows a VS Code editor with a Python file named 'AAC A 12.3.py'. The code defines functions for generating an inventory, searching by ID (binary search), searching by name (hash map), and sorting by price (quick sort). The terminal output shows the execution of these functions, including the generation of an inventory, binary search for a product with ID 5000, hash map search for a product with name 'Product_5000', and sorting the inventory by price. The output also includes a table mapping operation to algorithm to justification.

```

1 import random, time
2
3 def generate_inventory(n):
4     return [{"id": i, "name": f"Product_{i}", "price": round(random.uniform(10, 5000), 2),
5             "qty": random.randint(0, 500)} for i in range(1, n + 1)]
6
7 def binary_search_by_id(products, target_id):
8     lo, hi = 0, len(products) - 1
9     while lo <= hi:
10         mid = (lo + hi) // 2
11         if products[mid]["id"] == target_id:
12             return products[mid]
13         elif products[mid]["id"] < target_id:
14             lo = mid + 1
15         else:
16             hi = mid - 1
17     return None
18
19 def hash_search_by_name(index, name):
20     return index.get(name)
21
22 def build_name_index(products):
23     return {p["name"]: p for p in products}
24
25
26 PS C:\Users\Public> cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.8.win32-x64\debugpy\launcher' '57500' '-.' 'c:\Users\Public\AAC A 12.3.py'
== Search by ID (Binary Search) ==
Found: {'id': 5000, 'name': 'Product_5000', 'price': 2625.52, 'qty': 428} Time: 0.000007s
== Search by name (Hash Map) ==
Found: {'id': 5000, 'name': 'Product_5000', 'price': 2625.52, 'qty': 428} Time: 0.000002s
== Sort by Price (Quick sort) ==
Time: 0.00000s
ID Name Price Qty

```

Operation	Algorithm	Justification
Search by ID	Binary Search	$O(\log n)$ — IDs are sequential/sorted, minimal comparisons
Search by Name	Hash Map (dict)	$O(1)$ average — instant lookup, ideal for frequent name queries
Sort by Price/Qty	Quick Sort	$O(n \log n)$ average — fast in practice, good cache locality

Why these choices for inventory:

Describe what to build next

WelcomeAAC A 12.3.py X

C: > Users > Public > AAC A 12.3.py > ...

```
22 def build_name_index(products):
23     return {p["name"]: p for p in products}
24
25 def quick_sort(arr, key):
26     if len(arr) <= 1:
27         return arr
28     pivot = arr[len(arr) // 2][key]
29     left = [x for x in arr if x[key] < pivot]
30     mid = [x for x in arr if x[key] == pivot]
31     right = [x for x in arr if x[key] > pivot]
32     return quick_sort(left, key) + mid + quick_sort(right, key)
33
34 def display(products, n=10):
35     print(f"{'ID':<8}{'Name':<16}{'Price':<12}{'Qty':<6}")
36     print("-" * 42)
37     for p in products[:n]:
38         print(f"{p['id']:<8}{p['name']:<16}{p['price']:<12}{p['qty']:<6}")
39
40 inventory = generate_inventory(10000)
41
42 print("=== Search by ID (Binary Search) ===")
43 sorted_by_id = sorted(inventory, key=lambda x: x["id"])
44 start = time.perf_counter()
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

```
PS C:\Users\Public> c::; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Publ
```

ID	Name	Price	Qty
3623	Product_3623	10.64	206
9538	Product_9538	10.67	14
122	Product_122	11.15	333
2235	Product_2235	11.41	386
6376	Product_6376	11.61	320
5937	Product_5937	11.75	324
3876	Product_3876	11.77	356

Welcome

AAC A 12.3.py X

C: > Users > Public > AAC A 12.3.py > ...

```
42 print("=== Search by ID (Binary Search) ===")
43 sorted_by_id = sorted(inventory, key=lambda x: x["id"])
44 start = time.perf_counter()
45 result = binary_search_by_id(sorted_by_id, 5000)
46 t1 = time.perf_counter() - start
47 print(f"Found: {result} Time: {t1:.6f}s")
48
49 print("\n=== Search by Name (Hash Map) ===")
50 name_index = build_name_index(inventory)
51 start = time.perf_counter()
52 result = hash_search_by_name(name_index, "Product_5000")
53 t2 = time.perf_counter() - start
54 print(f"Found: {result} Time: {t2:.6f}s")
55
56 print("\n=== Sort by Price (Quick Sort) ===")
57 start = time.perf_counter()
58 sorted_price = quick_sort(inventory, "price")
59 t3 = time.perf_counter() - start
60 print(f"Time: {t3:.4f}s")
61 display(sorted_price)
62
63 print("\n=== Sort by Quantity (Quick Sort) ===")
64 start = time.perf_counter()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\Public> c:;; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda\
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920'
```

122	Product_122	11.15	333
2235	Product_2235	11.41	386
6376	Product_6376	11.61	320
5937	Product_5937	11.75	324
3876	Product_3876	11.77	356
6326	Product_6326	12.62	235
8471	Product_8471	13.08	105
2856	Product_2856	13.3	141

Welcome AAC A 12.3.py X

C: > Users > Public > AAC A 12.3.py > ...

```
57 start = time.perf_counter()
58 sorted_price = quick_sort(inventory, "price")
59 t3 = time.perf_counter() - start
60 print(f"Time: {t3:.4f}s")
61 display(sorted_price)
62
63 print("\n=== Sort by Quantity (Quick Sort) ===")
64 start = time.perf_counter()
65 sorted_qty = quick_sort(inventory, "qty")
66 t4 = time.perf_counter() - start
67 print(f"Time: {t4:.4f}s")
68 display(sorted_qty)
69
70 print("\n=== Algorithm Recommendation Table ===")
71 print(f'{"Operation":<25}{"Algorithm":<20}{"Justification"}')
72 print("-" * 75)
73 print(f'{"Search by ID":<25}{"Binary Search":<20}{"O(log n), data sorted by ID"}')
74 print(f'{"Search by Name":<25}{"Hash Map (dict)":<20}{"O(1) average lookup"}')
75 print(f'{"Sort by Price/Qty":<25}{"Quick Sort":<20}{"O(n log n) avg, fast in practice"}')
76
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c:; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\Public\AAC A 12.3.py'

=== Sort by Quantity (Quick Sort) ===
Time: 0.0557s

ID	Name	Price	Qty
85	Product_85	4139.68	0
441	Product_441	4613.7	0
1338	Product_1338	3868.26	0
1408	Product_1408	56.08	0

Welcome AAC A 12.3.py X

C:\Users\Public> AAC A 12.3.py > ...
start = time.perf_counter()
58 sorted_price = quick_sort(inventory, "price")
59 t3 = time.perf_counter() - start
60 print(f"Time: {t3:.4f}s")
61 display(sorted_price)
62
63 print("\n=== Sort by Quantity (Quick Sort) ===")
64 start = time.perf_counter()
65 sorted_qty = quick_sort(inventory, "qty")
66 t4 = time.perf_counter() - start
67 print(f"Time: {t4:.4f}s")
68 display(sorted_qty)
69
70 print("\n=== Algorithm Recommendation Table ===")
71 print(f"{'Operation':<25}{'Algorithm':<20}{'Justification'}")
72 print("-" * 75)
73 print(f"{'Search by ID':<25}{'Binary Search':<20}{'O(log n), data sorted by ID'}")
74 print(f"{'Search by Name':<25}{'Hash Map (dict)':<20}{'O(1) average lookup'}")
75 print(f"{'Sort by Price/Qty':<25}{'Quick Sort':<20}{'O(n log n) avg, fast in practice'}")
76

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c::; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Public\AAC A 12
441 Product_441 4613.7 0
1338 Product_1338 3868.26 0
1408 Product_1408 56.08 0
1740 Product_1740 660.14 0
2077 Product_2077 2768.81 0
2597 Product_2597 299.16 0
2837 Product_2837 3044.18 0
3438 Product_3438 3553.94 0
3474 Product_3474 3410.59 0

```
C:\Users\Public> AAC A 12.3.py > ...
57 start = time.perf_counter()
58 sorted_price = quick_sort(inventory, "price")
59 t3 = time.perf_counter() - start
60 print(f"Time: {t3:.4f}s")
61 display(sorted_price)
62
63 print("\n=== Sort by Quantity (Quick Sort) ===")
64 start = time.perf_counter()
65 sorted_qty = quick_sort(inventory, "qty")
66 t4 = time.perf_counter() - start
67 print(f"Time: {t4:.4f}s")
68 display(sorted_qty)
69
70 print("\n=== Algorithm Recommendation Table ===")
71 print(f"{'Operation':<25}{'Algorithm':<20}{'Justification'}")
72 print("-" * 75)
73 print(f"{'Search by ID':<25}{'Binary Search':<20}{'O(log n), data sorted by ID'}")
74 print(f"{'Search by Name':<25}{'Hash Map (dict)':<20}{'O(1) average lookup'}")
75 print(f"{'Sort by Price/Qty':<25}{'Quick Sort':<20}{'O(n log n) avg, fast in practice'}")
76
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Public> c;; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' '
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57920' '--' 'c:\Users\Public\AAC A 12.3
3474 Product_3474 3410.59 0

=== Algorithm Recommendation Table ===
Operation                Algorithm                Justification
-----
Search by ID              Binary Search            O(log n), data sorted by ID
Search by Name            Hash Map (dict)          O(1) average lookup
Sort by Price/Qty         Quick Sort               O(n log n) avg, fast in practice
PS C:\Users\Public>
```

Task 5: Real-Time Stock Data Sorting & Searching

Scenario:

An AI-powered **FinTech Lab** at SR University is building a tool for analyzing **stock price movements**. The requirement is to quickly **sort stocks by daily gain/loss** and search for specific stock symbols efficiently.

- Use **GitHub Copilot** to fetch or simulate stock price data (Stock Symbol, Opening Price, Closing Price).
- Implement sorting algorithms to rank stocks by **percentage change**.
- Implement a **search function** that retrieves stock data instantly when a stock symbol is entered.
- Optimize sorting with **Heap Sort** and searching with **Hash Maps**.
- Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.

WelcomeAAC A 12.3.py X

C:\Users> Public > AAC A 12.3.py > ...
1 import random, time, heapq
2
3 symbols = ["AAPL", "GOOG", "MSFT", "AMZN", "ISLA", "META", "NFLX", "NVDA", "INCY", "TCS",
4 "RELIANCE", "HDFC", "WIPRO", "KCLTECH", "SBIN", "LICICI", "BAJAJ", "SUNPHARMA",
5 "LT", "MARUTI", "TATAMOTORS", "ITC", "AXISBANK", "KOTAK", "ADANI"]
6
7 def generate_stocks(n):
8 stocks = []
9 for i in range(n):
10 sym = symbols[i % len(symbols)] + (f"{i//25}" if i >= 25 else "")
11 open_p = round(random.uniform(100, 5000), 2)
12 close_p = round(open_p * random.uniform(0.9, 1.1), 2)
13 pct = round((close_p - open_p) / open_p * 100, 2)
14 stocks.append({"symbol": sym, "open": open_p, "close": close_p, "pct_change": pct})
15 return stocks
16
17 def heap_sort_by_pct(stocks):
18 heap = [(-s["pct_change"], i, s) for i, s in enumerate(stocks)]
19 heapq.heapify(heap)
20 return [heapq.heappop(heap)[2] for _ in range(len(heap))]
21
22 def hash_search(index, symbol):
23 return index.get(symbol)
24

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS
PS C:\Users\Public> c:\cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-
thon.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58347' '-.' 'c:\Users\Public\AAC A 12.3.py'
=== Heap Sort vs sorted() - Rank by % Change ===

Heap Sort : 0.0091s
sorted() : 0.0018s
Faster : sorted()

Top 10 Gainers:
SymbolOpenCloseChange %

NFLX_1192147.642362.279.99
AXISBANK_149313.62344.949.99
META_235160.33175.349.99

CHAT
SORTING STUDENT RECORDS FOR PL...
stock price data (Stock Symbol, Opening Price, Closing Price).
• Implement sorting algorithms to rank stocks by percentage change.
• Implement a search function that retrieves stock data instantly when a stock symbol is entered.
• Optimize sorting with Heap Sort and searching with Hash Maps.
• Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.
give simple and small code in python
✓ Reviewed AAC A 12.3.py
✓ Created AAC A 12.3.py
python
✓ "c:\Users\Public\AAC A 12.3.py"
Code written to AAC A 12.3.py. Results:
Sorting (10,000 stocks by % change):
• Heap Sort: 0.0056s
• sorted() (Timsort): 0.0030s — faster because Timsort is a highly optimized hybrid algorithm
AAC A 12.3.py
Describe what to build next
Claude Opus 4.5

WelcomeAAC A 12.3.py X

C:\Users> Public > AAC A 12.3.py > ...
28 def display(stocks, n=10):
32 print(f"{s['symbol']:<15}{s['open']:<12}{s['close']:<12}{s['pct_change']:<10}")
33
34 data = generate_stocks(10000)
35
36 print("=== Heap Sort vs sorted() - Rank by % Change ===\n")
37
38 start = time.perf_counter()
39 heap_sorted = heap_sort_by_pct(data)
40 t_heap = time.perf_counter() - start
41
42 start = time.perf_counter()
43 lib_sorted = sorted(data, key=lambda x: -x["pct_change"])
44 t_lib = time.perf_counter() - start
45
46 print(f"Heap Sort : {t_heap:.4f}s")
47 print(f"sorted() : {t_lib:.4f}s")
48 print(f"Faster : {'Heap Sort' if t_heap < t_lib else 'sorted()'}\n")
49
50 print("Top 10 Gainers:")
51 display(heap_sorted)
52 print("\nTop 10 Losers:")
53 display(heap_sorted[-10:][::-1])
54

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS
PS C:\Users\Public> c:\cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\m
thon.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58347' '-.' 'c:\Users\Public\AAC A 12.3.py'

NFLX_1192147.642362.279.99
AXISBANK_149313.62344.949.99
META_235160.33175.349.99

Welcome AAC A 12.3.py X

C:\> Users > Public > AAC A 12.3.py > ...

```
28 def display(stocks, n=10):
32     print(f"{s['symbol']:<15}{s['open']:<12}{s['close']:<12}{s['pct_change']:<10}")
33
34     data = generate_stocks(10000)
35
36     print("=== Heap Sort vs sorted() - Rank by % Change ===\n")
37
38     start = time.perf_counter()
39     heap_sorted = heap_sort_by_pct(data)
40     t_heap = time.perf_counter() - start
41
42     start = time.perf_counter()
43     lib_sorted = sorted(data, key=lambda x: -x["pct_change"])
44     t_lib = time.perf_counter() - start
45
46     print(f"Heap Sort : {t_heap:.4f}s")
47     print(f"sorted() : {t_lib:.4f}s")
48     print(f"Faster : {'Heap Sort' if t_heap < t_lib else 'sorted()'}\n")
49
50     print("Top 10 Gainers:")
51     display(heap_sorted)
52     print("\nTop 10 Losers:")
53     display(heap_sorted[-10:][::-1])
54
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c;; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher '58347' '--' 'c:\Users\Public\AAC A 12

NFLX_119	2147.64	2362.27	9.99
AXISBANK_149	313.62	344.94	9.99
META_225	160.32	176.34	9.99

Top 10 Losers:

Symbol	Open	Close	Change %
ITC_339	4372.67	3935.42	-10.0
BAJAJ_316	3547.91	3193.18	-10.0

VS Code Window: AAC A 12.3.py X

C:\Users\Public> AAC A 12.3.py > ...

```
28 def display(stocks, n=10):
32     print(f"{s['symbol']:<15}{s['open']:<12}{s['close']:<12}{s['pct_change']:<10}")
33
34     data = generate_stocks(10000)
35
36     print("=== Heap Sort vs sorted() - Rank by % Change ===\n")
37
38     start = time.perf_counter()
39     heap_sorted = heap_sort_by_pct(data)
40     t_heap = time.perf_counter() - start
41
42     start = time.perf_counter()
43     lib_sorted = sorted(data, key=lambda x: -x["pct_change"])
44     t_lib = time.perf_counter() - start
45
46     print(f"Heap Sort : {t_heap:.4f}s")
47     print(f"sorted() : {t_lib:.4f}s")
48     print(f"Faster : {'Heap Sort' if t_heap < t_lib else 'sorted()'}\n")
49
50     print("Top 10 Gainers:")
51     display(heap_sorted)
52     print("\nTop 10 Losers:")
53     display(heap_sorted[-10:][::-1])
54
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c:: cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.

ADANI_258	3536.14	3182.58	-10.0
HCLTECH_254	3068.46	2761.7	-10.0
SUNPHARMA_222	3697.36	3327.72	-10.0
MARUTI_93	1783.28	1604.99	-10.0
WIPRO_334	861.71	775.65	-9.99
ITC_327	2201.07	1981.19	-9.99
GOOG_285	2888.79	2600.15	-9.99
NFLX_256	2957.21	2661.79	-9.99

```
Welcome AAC A 12.3.py X
C: > Users > Public > AAC A 12.3.py > ...
49
50 print("Top 10 Gainers:")
51 display(heap_sorted)
52 print("\nTop 10 Losers:")
53 display(heap_sorted[-10:][::-1])
54
55 print("\n=== Hash Map vs dict lookup - Search by Symbol ===\n")
56
57 index = build_hash_index(data)
58
59 start = time.perf_counter()
60 r1 = hash_search(index, "TSLA")
61 t_hash = time.perf_counter() - start
62
63 start = time.perf_counter()
64 r2 = next((s for s in data if s["symbol"] == "TSLA"), None)
65 t_linear = time.perf_counter() - start
66
67 print(f"Hash Map Search : {t_hash:.6f}s Result: {r1}")
68 print(f"Linear Search : {t_linear:.6f}s Result: {r2}")
69 print(f"Faster : {'Hash Map' if t_hash < t_linear else 'Linear'}")
70
71 print("\n=== Performance Comparison Table ===")
72 print(f"{'Operation':<20}{ 'Custom':<18}{ 'Standard Lib':<18}{ 'Trade-off'}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c;; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'

=== Hash Map vs dict lookup - Search by Symbol ===

Hash Map Search : 0.000003s Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Linear Search : 0.000013s Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Faster : Hash Map

=== Performance Comparison Table ===
Operation Custom Standard Lib Trade-off
----- Start -----
```

```
Welcome AAC A 12.3.py X
C: > Users > Public > AAC A 12.3.py > ...
60 r1 = hash_search(index, "TSLA")
61 t_hash = time.perf_counter() - start
62
63 start = time.perf_counter()
64 r2 = next((s for s in data if s["symbol"] == "TSLA"), None)
65 t_linear = time.perf_counter() - start
66
67 print(f"Hash Map Search : {t_hash:.6f}s Result: {r1}")
68 print(f"Linear Search : {t_linear:.6f}s Result: {r2}")
69 print(f"Faster : {'Hash Map' if t_hash < t_linear else 'Linear'}")
70
71 print("\n=== Performance Comparison Table ===")
72 print(f"{'Operation':<20}{ 'Custom':<18}{ 'Standard Lib':<18}{ 'Trade-off'}")
73 print(f"{'-' * 75}")
74 print(f"{'Sort by % change':<20}{ 'Heap Sort':<18}{ 'sorted()':<18}{ 'Timsort is hybrid, often faster'}")
75 print(f"{'Search by symbol':<20}{ 'Hash Map O(1)':<18}{ 'Linear O(n)':<18}{ 'Hash uses more memory'}")
76

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Public> c;; cd 'c:\Users\Public'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\thon.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58347' '--' 'c:\Users\Public\AAC A 12.3.py'

Linear Search : 0.000013s Result: {'symbol': 'TSLA', 'open': 3265.84, 'close': 2963.07, 'pct_change': -9.27}
Faster : Hash Map

=== Performance Comparison Table ===
Operation Custom Standard Lib Trade-off
----- Start -----
Sort by % change Heap Sort sorted() Timsort is hybrid, often faster
Search by symbol Hash Map O(1) Linear O(n) Hash uses more memory
PS C:\Users\Public>
```


	Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	
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